

The diagram shows a curved path labeled "Path". A robot is represented by an oval at position (x, y) with a heading θ . A target point is marked with a purple dot at (x_f, y_f) . A red line segment labeled e connects the robot to the "nearest point" on the path. A black arrow labeled v_f points from the robot towards the target. Angles δ and δ_c are shown in green, representing the deviation from the path's tangent. A blue angle θ_c is also indicated. The distance from the robot to the target along the path is labeled L .